Assignment for the Course "Causal Machine Learning"

Format of the assignment
This assignment is a group project. Students are expected to work in groups of 2–3 students. Each group will work on one empirical project and submit a short term paper of a maximum of 6 pages, including figures, tables, and references. In addition, each group will give a 15-minutes presentation in class on the last day of the course.

Grading
The final grade is composed as follows:

- Written group assignment (term paper): **50%**
- Group presentation (last course day): **25%**
- Participation during the course: **25%**

Empirical background
Low voter turnout can be a problem for the legitimisation of political representation. One common intervention to increase turnout consists of so-called get-out-the-vote (GOTV) calls, which provide information about the election date and the importance of voting. In 2002, an experiment was conducted in Michigan and Iowa to study the effect of GOTV calls on voter turnout.[1] Voters were randomly assigned to treatment and control groups. One group of approximately 60,000 voters was assigned to receive a GOTV call, although not all of them were successfully reached. The control group, consisting of nearly 2 million voters, did not receive a call.

[1] Arceneaux, K., Gerber, A. S. and Green, D. P. (2006). *Comparing Experimental and Matching Methods Using a Large-Scale Voter Mobilization Experiment.* Political Analysis, 14(1), 37–62.

Data
The GitHub folder contains the dataset `mobilization.RData` and `mobilization.csv` with the following variables:

- *vote02*: voted in 2002 (0 = no, 1 = yes)
- *treat_real*: received a GOTV call (1 = yes, 0 = no)
- *contact*: answered a GOTV call (1 = yes, 0 = no)
- *persons*: number of persons in the household
- *newreg*: newly registered voter
- *age*: age in years
- *female*: gender (1 = female, 0 = male)
- *vote00*: voted in 2000
- *vote98*: voted in 1998
- *state*: 0 = Michigan, 1 = Iowa
- *competiv*: district competitiveness
- *comp_mi*: competitiveness × Michigan
- *comp_ia*: competitiveness × Iowa

Available projects
Each group must choose **one** of the following two projects:

1. Use a double/debiased machine learning approach to estimate the causal effect of answering a GOTV call on voter turnout in the 2002 election.
2. Conduct a machine-learning-based heterogeneity analysis of the effect of receiving a GOTV call on voter turnout in the 2002 election.

General questions
Regardless of the chosen project, the following aspects should be addressed in the paper:

- Data quality: Are there data issues or inconsistencies?

- Identification strategy: What assumptions are required, and how plausible are they?
- Methodological discussion: What are the strengths and limitations of the chosen approach?